

Kaniadakis Information Geometry of Compositional Data

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Abstract: A special case of Kaniadakis' logarithm has been independently proposed for analyzing compositional data in the Atchison style. We expand on this finding by showing that the affine information geometry derived from the κ -logarithm provides a consistent setup for the geometric analysis of compositional data.

Keywords: Kaniadakis logarithm; Information geometry; Compositional data; Affine displacement; Affine statistical bundle; Barycenter; Kaniadakis divergence

1. Introduction

This paper describes Kaniadakis' statistics as a methodology in data science. We present the methods in some generality and use them for the exploratory analysis of compositional data. The illustrating example is a small data set, and we do not discuss any scaling issues of our methods. However, the sample has an independent interest in financial risk analysis.

1.1. Why a geometric methodology

Kaniadakis' logarithm Kaniadakis [1,2] provides a generalization of the ordinary logarithm that can support full generalized development of exponential families, statistical divergences, and entropy. More specifically, it supports the presentation of the geometry of the probability simplex as a system of two affine spaces in duality as it is in Information Geometry IG Amari and Nagaoka [3]. The systematic use of this formal geometric perspective provides a robust and unified rationale for discussing key descriptive concepts. Notable examples are the barycenter, the displacement from the barycenter, and the dimensionality reduction. In the affine setting we propose below, the definition of such notions above follows uniquely from the general principle. For the affine geometry of the probability simplex, see, for example, the tutorial reference Chirco and Pistone [4]. We use a special kind of Kaniadakis' logarithm that appeared with a different name in Compositional Data (CoDa) [5, Example 4.20].

1.2. CoDa

CoDa is the data where all of a (row) vector's (i.e. $[x_1, x_2, \dots, x_D]$) components are strictly positive real values and contain solely relative information; the composition is called a D-part composition. Compositional data is often expressed as closed form and total to a fixed value, such as 1 for parts per unit or 100 for percentage measurements [6, Chapter 2].

Compositional data is often found in geosciences and other scientific disciplines for classification, discrimination, and categorization. CoDa analysis is closely related to geosciences and biology, where the data is mostly expressed as proportions or concentrations without mentioning the total size explicitly Egozcue and Pawłowsky-Glahn [7]. Significant advancement has been accomplished during the last thirty to forty years, and recently the term CoDa analysis has been employed to "Insist on the idea that the study goals or hypotheses, which place more of an emphasis on relative than absolute values, are what

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ultimately determine composition rather than the data, which may not be pieces of a whole or may not have a fixed sum" Coenders and Ferrer-Rosell [8]. These qualities make CoDa analysis the most powerful tool for applications outside the tradition of hard sciences Fiori and Porro [9]. Current studies in management, economics, and social sciences have shown in practice the benefits of compositional methods in handling a wide variety of problems, which range from market shares, and customer segmentation to tourism, transport systems, financial ratios, and many more (see Coenders and Ferrer-Rosell [8], Linares-Mustarós et al. [10], Boonen et al. [11], Grifoll et al. [12]).

1.3. CoDa and systemic financial risk

Atchison Aitchison [13] first introduced CoDa analysis. The research conducted by the Fiori and Porro [9] on financial data used the Atchison methods to examine how European nations contribute to the total amount of systemic risk (SRISK). They find that the distinctive quality of CoDa analysis, especially the Atchison geometry, is very effective in determining the threats of possible instability concerns offered by smaller institutions and nations that might not completely emerge from the scale of their systemic risk. CoDa analysis, in addition, when compared to previous literature Stolbov and Shchepeleva [14], Dzhagityan and Orekhov [15] they used size of systemic risk (SRISK) using either financial units: popular indicators expressed in financial units include, among others, SRISK and Covariance (CoVaR) based measures for comparative analysis.

The Center for Risk Management at the University of Lausanne provides systemic risk assessments for European financial institutions, which we used in our empirical study using the above Kaniadakis methods (<http://www.crml.ch>). The dataset enables the determination of SRISK country-level values, a market-based systemic risk indicator first proposed in Acharya et al. [16,17] and most recently examined in Engle [18].

The characteristics of SRISK are popular in the literature, and mainly SRISK is used to recognize weak institutions and countries with a system-wide impact before a crisis occurs Stolbov and Shchepeleva [14] and can help forecast actual sector performance Engle et al. [19].

Most of the previous literature has mainly focused on the absolute values of SRISK. In this work, we focus on implementing the Kaniadakis methods to see the different European countries as a part of compositional data. We developed work started in Fiori and Porro [9], where they first introduced compositional data analysis to examine the distribution of relative contributions to SRISK connected with key European nations from 2008 to 2021.

1.4. Data and methods

This paper first establishes a novel theoretical framework for compositional data using Kaniadakis logarithm. Second, implementing Kaniadakis divergence on the compositional data, calculating the exponential and mixture displacement on compositional data. Later, we calculate the barycenter and deviation. The purpose of calculating the barycenter is to check how far are values of SRISK are from their center value.

In this research, we take the sample of ten European economies (Belgium, Denmark, France, Germany, Greece, Italy, Netherlands, Spain, Switzerland, UK) with annual SRISK measurements collected at the end of December for 2008–2021. Every number is stated in billions of euros. Like most CoDa method applications, the sample does not cover Europe. Therefore, the 10 components that make up our SRISK compositions are just a portion of all those that may be used. CoDa analysis, however, is predicated on the basic notion of sub-compositional coherence, which ensures that a compositional study conducted on a subset of components is consistent with the same analysis performed on the entire composition.

1.5. Recap: Kaniadakis' logarithm

We focus on the special case of Kaniadakis' logarithm with a purely algebraic form. In the suggestive formalism introduced by Naudts [20] the generalised logarithms are associated to the reciprocal derivative function A ,

$$\log_{\kappa}(x) = \frac{1}{2} \left(x - \frac{1}{x} \right) = \int_1^x \frac{du}{A(u)}, \quad \text{with} \quad A(u) = \frac{2u^2}{1+u^2}, \quad (1)$$

$$\exp_{\kappa}(y) = \log_{\kappa}^{-1} y = y + \sqrt{1+y^2} = \exp \int_0^y \frac{dv}{\sqrt{1+v^2}}, \quad x^2 - 2xy - 1 = 0, \quad x > 0. \quad (2)$$

Notice that the growth is linear in both directions. Notice also that the last equation above reduces any polynomial in y and $x = \exp_{\kappa}(y)$, for example,

$$\begin{aligned} \exp_{\kappa}(y)^2 &= 2y \exp_{\kappa}(y) + 1, \\ \exp_{\kappa}(y)^3 &= \exp_{\kappa}(y)(2y \exp_{\kappa}(y) + 1) = \dots, \end{aligned}$$

and so on. This is an algebraic feature, and this theory is a case of algebraic statistics Pistone et al. [21].

We list below the main known properties of the κ -logarithm and the κ -exponential.

$$\log_{\kappa} \left(\frac{1}{x} \right) = -\log_{\kappa}(x), \quad \exp_{\kappa}(y) \exp_{\kappa}(-y) = 1. \quad (3)$$

$$\frac{d}{dy} \exp_{\kappa}(y) = A(\exp_{\kappa}(y)) = \frac{1}{\sqrt{1+y^2}} \exp_{\kappa}(y) \quad (4)$$

$$\exp_{\kappa}(x) \exp_{\kappa}(y) = \exp_{\kappa} \left(x \sqrt{1+y^2} + y \sqrt{1+x^2} \right) \quad (5)$$

$$\log_{\kappa}(x_2) - \log_{\kappa}(x_1) < \frac{1}{A(x_1)}(x_2 - x_1), \quad x_1 \neq x_2 \quad (6)$$

$$\exp_{\kappa}(y_2) - \exp_{\kappa}(y_1) > A(\exp_{\kappa}(y_1))(y_2 - y_1), \quad y_1 \neq y_2 \quad (7)$$

1.6. Kaniadakis' exponential form of a positive probability function

If Ω is a finite sample space, the probability simplex on Ω is $\mathcal{P}(\Omega)$ and the open probability simplex is $\mathcal{E}(\Omega)$.

For all $p \in \mathcal{E}(\Omega)$, the function $A \circ p$, A as in eq. (1), is strictly positive and provides a positive weight on Ω . It is proportional, but usually not equal, to a probability function,

$$\tilde{p} \propto A \circ p, \quad \text{namely,} \quad \tilde{p}(x) = \frac{A(p(x))}{\sum_{x \in \Omega} A(p(x))}.$$

We will also write $\tilde{p} = A \circ p / \overline{A \circ p}$. The mapping $\mathcal{E}(\Omega) \ni p \mapsto \tilde{p} \in \mathcal{E}(\Omega)$ is the *escort* mapping, see [20]. See [22, §3.1] for a discussion of its injectivity and surjectivity. We introduce a notation for the escort expectation, $\tilde{\mathbb{E}}_p[u] = \sum_{x \in \Omega} u(x) \tilde{p}(x)$.

For $p, q \in \mathcal{E}(\Omega)$, we define the *Kaniadakis divergence* as by changing in the usual definition the logarithm to the Kaniadakis logarithm and the probability function p with the escort $\tilde{p}$:

$$\begin{aligned} \tilde{D}(p|q) &= \tilde{\mathbb{E}}_p[\log_{\kappa}(p) - \log_{\kappa}(q)] \\ &= \sum_{x \in \Omega} (\log_{\kappa}(p) - \log_{\kappa}(q)) A \circ p / \overline{A \circ p}. \end{aligned}$$

$$= \sum_{x \in \Omega} (\log_{\kappa}(p) - \log_{\kappa}(q)) \tilde{p}(x) \quad (8)$$

Clearly, $\tilde{D}(p|p) = 0$. If $p \neq q$, from the concavity in eq. (6), 108

$$\begin{aligned} \tilde{D}(p|q) &= (\overline{A \circ p})^{-1} \sum_{x \in \Omega} A(p(x)) (\log_{\kappa}(p(x)) - \log_{\kappa}(q(x))) \\ &> (\overline{A \circ p})^{-1} \sum_{x \in \Omega} (p(x) - q(x)) \\ &= 0. \end{aligned}$$

Fix $p \in \mathcal{E}(\Omega)$. For all $q \in \mathcal{E}(\Omega)$, define 109

$$s_p(q) = (\log_{\kappa}(q) - \log_{\kappa}(p)) + \tilde{D}(p|q) = (\log_{\kappa}(q) - \log_{\kappa}(p)) - \tilde{\mathbb{E}}_p[(\log_{\kappa}(q) - \log_{\kappa}(p))],$$

then, for $u = s_p(q)$, 110

$$q = \exp_{\kappa}(u - \tilde{D}(p|q) + \log_{\kappa}(p)), \quad \tilde{\mathbb{E}}_p[u] = 0. \quad (9)$$

Conversely, for all $p \in \mathcal{E}(\Omega)$, if u is a random variable such that $\tilde{\mathbb{E}}_p[u] = 0$, the real 111
function 112

$$\mathbb{R}_+ \ni \psi \mapsto \sum_{x \in \Omega} \exp_{\kappa}(u(x) - \psi + \log_{\kappa}(p(x)))$$

is continuous, goes to 0 as $\psi \rightarrow \infty$, and, for $\psi = 0$, takes a value larger than 1 because of eq. (7):

$$\begin{aligned} \sum_{x \in \Omega} \exp_{\kappa}(u(x) + \log_{\kappa}(p(x))) &> \sum_{x \in \Omega} \exp_{\kappa}(\log_{\kappa}(p(x))) + \sum_{x \in \Omega} A(\exp_{\kappa}(\log_{\kappa}(p(x))))u(x) \\ &= \sum_{x \in \Omega} p(x) + \sum_{x \in \Omega} A(p(x))u(x) \\ &= 1. \end{aligned}$$

In conclusion, there exists a function, 113

$$K_p: S_p = \left\{ u \in L(\Omega) \mid \tilde{\mathbb{E}}_p[u] = 0 \right\} \ni u \mapsto K_p(u) \geq 0,$$

and $K_p(u) > 0$ provided $u \neq 0$, such that 114

$$q = \exp_{\kappa}(u - K_p(u) + \log_{\kappa}(p)) \in \mathcal{E}(\Omega),$$

hence $K_p(u) = \tilde{D}(p|q)$. The mapping

$$s_p: \mathcal{E}(\Omega) \rightarrow S_p$$

is a bijection with inverse 115

$$e_p: S_p \ni u \mapsto \exp_{\kappa}(u - K_p(u) + \log_{\kappa}(p)) \in \mathcal{E}(\Omega). \quad (10)$$

1.7. Properties of the cumulant function K_p . 116

Let us compute the derivatives of the function K_p . We use a square bracket notation 117
for the direction, 118

$$dK_p(u)[h] = \lim_{\theta \rightarrow 0} \theta^{-1} (K_p(u + \theta h) - K_p(u)) = \left. \frac{d}{d\theta} K_p(u + \theta h) \right|_{\theta=0}.$$

From eq. (10),

$$\begin{aligned}
 0 &= \frac{d}{d\theta} \sum_{x \in \Omega} \exp_{\kappa}(u(x) + \theta h(x) - K_p(u + \theta h) + \log_{\kappa}(p(x))) \Big|_{\theta=0} \\
 &= \sum_{x \in \Omega} A(\exp_{\kappa}(u(x) + \theta h(x) - K_p(u + \theta h) + \log_{\kappa}(p(x))))(h(x) - \frac{d}{dt} K_p(u + \theta h)) \Big|_{\theta=0} \\
 &= \sum_{x \in \Omega} A(\exp_{\kappa}(u(x) - K_p(u) + \log_{\kappa}(p(x))))(h(x) - dK_p(u)[h]) \\
 &= \sum_{x \in \Omega} A(e_p(u))(h - dK_p(u)[h]) .
 \end{aligned}$$

It follows that for each $p \in \mathcal{E}(\Omega)$, and $u, h \in S_p$, it holds

$$dK_p(u)[h] = \tilde{\mathbb{E}}_q[h] , \quad \text{where } q = e_p(u) \text{ see eq. (10)} . \quad (11)$$

1.8. Bibliographical notes

Similarly, one should compute $d^2 K_p$ and the convex conjugate of K_p . See below for the duality and see also Pistone [23], Chirco and Pistone [4]. Kaniadakis logarithm and exponential were first introduced in Kaniadakis [24]; Kaniadakis [25]. The application to IG used here appeared in Pistone [26]; Montrucchio and Pistone [27]; Montrucchio and Pistone [22]. These papers discuss both the finite state space and the general state space.

2. Affine space

The Kaniadakis non-parametric affine geometry of the open probability simplex is a variation of the standard case Chirco and Pistone [4]. The main difference is the substitution of the expectation with the escort expectation.

2.1. Statistical bundle

The statistical bundle is an expression of the tangent space of $\mathcal{E}(\Omega)$ as a dually flat affine statistical manifold in the sense of Amari and Nagaoka [3]. The statistical bundle $S\mathcal{E}(\Omega)$ and each fiber $S_q\mathcal{E}(\Omega)$ are defined by

$$S\mathcal{E}(\Omega) = \left\{ (q, v) \mid q \in \mathcal{E}(\Omega), \tilde{\mathbb{E}}_q[v] = 0 \right\} , \quad (12)$$

$$S_q\mathcal{E}(\Omega) = \left\{ v \in L(\Omega) \mid \tilde{\mathbb{E}}_q[v] = 0 \right\} , \quad q \in \mathcal{E}(\Omega) . \quad (13)$$

In our setup, each fibre is a finite-dimensional vector space and can be identified with its dual. However, it is convenient to distinguish the two statistical bundles. The previous one is called *exponential* statistical bundle, while the *mixture* statistical bundle is

$${}^*S\mathcal{E}(\Omega) = \left\{ (q, v) \mid q \in \mathcal{E}(\Omega), \tilde{\mathbb{E}}_q[v] = 0 \right\} , \quad (14)$$

$${}^*S_q\mathcal{E}(\Omega) = \left\{ v \in L(\Omega) \mid \tilde{\mathbb{E}}_q[v] = 0 \right\} , \quad q \in \mathcal{E}(\Omega) . \quad (15)$$

For each couple $p, q \in \mathcal{E}(\Omega)$, the mapping

$${}^e\mathbb{U}_p^q: S_p\mathcal{E}(\Omega) \ni v \mapsto v - \tilde{\mathbb{E}}_q[v] \in S_q\mathcal{E}(\Omega) \quad (16)$$

is a bijection. The ${}^e\mathbb{U}_p^p$ is the identity mapping, and

$${}^e\mathbb{U}_q^r {}^e\mathbb{U}_p^q = {}^e\mathbb{U}_p^r ,$$

The co-cycle of mappings $({}^e\mathbb{U}_p^q)_{p,q}$ is the *exponential parallel transport* of the exponential statistical bundle.

The mapping defined for all $p \in \mathcal{E}(\Omega)$, $v \in {}^*S_p \mathcal{E}(\Omega)$, and $w \in S_p \mathcal{E}(\Omega)$ by

$$g: (p, v, w) \mapsto g_p(v, w) = \langle v, w \rangle_{A \circ p} = \sum_{x \in \Omega} A(p(x)) u(x) v(x) = \overline{A \circ p} \tilde{\mathbb{E}}_p[vw]$$

provides a duality between the fibres of $S \mathcal{E}(\Omega)$ and ${}^*S \mathcal{E}(\Omega)$.

Let us compute the dual of the exponential transport. For $p, q \in \mathcal{E}(\Omega)$, $v \in {}^*S_q \mathcal{E}(\Omega)$, and $w \in S_p \mathcal{E}(\Omega)$,

$$\begin{aligned} \left\langle v, {}^e\mathbb{U}_p^q w \right\rangle_{A \circ q} &= \sum_{x \in \Omega} A \circ q v(w - \tilde{\mathbb{E}}_q[w]) \\ &= \sum_{x \in \Omega} A \circ q v w - \tilde{\mathbb{E}}_q[w] \sum A \circ q v \\ &= \sum_{x \in \Omega} A \circ q v w \\ &= \sum_{x \in \Omega} A \circ p \left(\frac{A \circ q}{A \circ p} u \right) w \\ &= \left\langle \frac{A \circ q}{A \circ p} u, w \right\rangle_{A \circ p}. \end{aligned}$$

Now, $\frac{A \circ q}{A \circ p} v \in {}^*S_p \mathcal{E}(\Omega)$, hence the dual of the exponential transport is the *mixture transport*,

$${}^*\left({}^e\mathbb{U}_q^p\right)v = {}^m\mathbb{U}_q^p v = \frac{A \circ q}{A \circ p} v. \quad (17)$$

2.2. Velocity and auto-parallel curves

The following computation is a version of the original argument about Fisher's score. Let $t \mapsto q(t) \in \mathcal{E}(\Omega)$ be a one-dimensional parametric model, a curve in geometric language. We assume it is smooth as a mapping in the vector space $L(\Omega)$. For each random variable $f \in L(\Omega)$,

$$\begin{aligned} \frac{d}{dt} \mathbb{E}_{q(t)}[f] &= \sum_{x \in \Omega} f(x) \dot{q}(x; t) = \sum_{x \in \Omega} f(x) \frac{\dot{q}(x; t)}{A(q(x; t))} A(q(x; t)) = \\ &= \sum_{x \in \Omega} f(x) \frac{d}{dt} \log_{\kappa}(q(x; t) A(q(x; t))) = \sum_{x \in \Omega} A \circ q(t) (f - \mathbb{E}_{\tilde{q}(t)}[f]) \frac{d}{dt} \log_{\kappa}(q(t)) = \\ &= \left\langle f - \mathbb{E}_{\tilde{q}(t)}[f], \frac{d}{dt} \log_{\kappa}(q(t)) \right\rangle_{A \circ q(t)}. \end{aligned} \quad (18)$$

If we define the *velocity* of the curve to be

$$\dot{q}(t) = \frac{d}{dt} \log_{\kappa}(q(t)) = \frac{\dot{q}(t)}{A(q(t))}. \quad (19)$$

we can check that $\dot{q}(t) \in S_{q(t)} \mathcal{E}(\Omega)$ and $(f - \mathbb{E}_{\tilde{q}(t)}[f]) \in {}^*S_{q(t)} \mathcal{E}(\Omega)$. And symmetrically.

The *Cramer-Rao bound* is

$$\begin{aligned} \left(\frac{d}{dt} \mathbb{E}_{q(t)}[f] \right)^2 &= \overline{A \circ q(t)}^2 \left\langle f - \mathbb{E}_{\tilde{q}(t)}[f], \frac{d}{dt} \log_{\kappa}(q(t)) \right\rangle_{\tilde{q}(t)} \leq \\ &\overline{A \circ q(t)}^2 \mathbb{E}_{\tilde{q}(t)} \left[(f - \mathbb{E}_{\tilde{q}(t)}[f])^2 \right] \mathbb{E}_{\tilde{q}(t)} \left[\left(\frac{\dot{q}(t)}{A \circ q(t)} \right)^2 \right] = \\ &\sum_{x \in \Omega} A \circ q(t) (f - \tilde{\mathbb{E}}_q[f])^2 \sum \frac{\dot{q}(t)^2}{A \circ q(t)}. \quad (20) \end{aligned}$$

Notice that the variation computed with the escort probability function, namely $f - \tilde{\mathbb{E}}_q[f]$ appears in eq. (18) as a gradient of the expectation $f \mapsto \mathbb{E}_q[f]$. 155
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A curve $t \mapsto q(t)$ is *auto-parallel* for the mixture transport if 157

$$\mathbb{m}\mathbb{U}_{q(t)}^{q(s)} \dot{q}(t) = \dot{q}(s).$$

For $q(0) = q_0$ and $q(1) = q_1$, 158

$$\frac{\dot{q}(0)}{A \circ q_0} = \frac{A \circ q(t)}{A \circ q_0} \frac{\dot{q}(t)}{A \circ q(t)} = \frac{1}{A \circ q_0} \dot{q}(t),$$

so that

$$q(t) = q_0 + t(q_1 - q_0).$$

The auto-parallel curves for the mixture transport are the ordinary geodesics of the simplex. 159

Let us compute the auto-parallel curves for the exponential transport, 160

$$\mathbb{e}\mathbb{U}_{q(t)}^{q(s)} \dot{q}(t) = \dot{q}(s).$$

For $q(0) = q_0$ and $q(1) = q_1$, 161

$$\dot{q}(0) = \dot{q}(t) - \tilde{\mathbb{E}}_{q_0}[\dot{q}(t)] = \frac{d}{dt} \log_{\kappa}(q(t)) - \frac{1}{A \circ q_0} \sum A \circ q_0 \frac{d}{dt} \log_{\kappa}(q(t)),$$

so that, for some function ψ 162

$$\log_{\kappa}(q(t)) = \log_{\kappa}(q_0) + t\dot{q}(0) - \psi(t).$$

Comparing with eq. (10), we have that the auto-parallel curve for the exponential transport is 163
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$$q(t) = \exp_{\kappa}(t\dot{q}(0) - K_{q_0}(t\dot{q}(0)) + \log_{\kappa}(q_0)).$$

As seen above, 165

$$\dot{q}(0) = \log_{\kappa}(q_1) - \log_{\kappa}(q_0) - \tilde{\mathbb{E}}_{q_0}[\log_{\kappa}(q_1) - \log_{\kappa}(q_0)] = \log_{\kappa}(q_0) + \tilde{D}(q_0|q_1).$$

2.3. Displacement 166

The machinery introduced above allows for the explicit definition of the affine structure as originally defined by Weyl [28]. A textbook on affine geometry is Berger [29, Ch. 2,3,9]. 167
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We define the following two (dual) displacements on the statistical bundle. The mixture displacement is 169
170

$$\eta_p(q) = \frac{q - p}{A \circ p}. \quad (21)$$

The exponential displacement is 171

$$s_p(q) = (\log_\kappa(q) - \log_\kappa(p)) - \tilde{\mathbb{E}}_p[(\log_\kappa(q) - \log_\kappa(p))] \quad (22)$$

Both displacements define affine coordinates in the statistical bundle. The easy proofs are the same as in the standard cases Chirco and Pistone [4]. Each displacement defines an atlas of charts on the affine bundle.

The availability of an affine bundle would allow for a coherent and straightforward definition of mechanical concepts: velocity, acceleration, Lagrangian, and Hamiltonian See Pistone [30] and Chirco et al. [31] for the standard case. In the present paper, we develop the application to Data Science, and we stress the notion of affine barycenter and the notion that a system of charts can be seen as a preprocessing of data to be followed by any method adapted to actual vector data.

2.4. Barycenter and deviation

Let $f_1, \dots, f_n$ be a sequence of (positive) compositional data points. We assume normalization to 1. Hence the data are points in the open probability simplex. The affine coordinates eq. (22) at a reference point p are

$$\begin{aligned} s_p(f_1) &= (\log_\kappa(f_1) - \log_\kappa(p)) - \tilde{\mathbb{E}}_p[(\log_\kappa(f_1) - \log_\kappa(p))] \\ &\vdots \\ s_p(f_n) &= (\log_\kappa(f_n) - \log_\kappa(p)) - \tilde{\mathbb{E}}_p[(\log_\kappa(f_n) - \log_\kappa(p))] \end{aligned}$$

The mean value is

$$\begin{aligned} \bar{s}_p &= \frac{1}{n} \sum_{j=1}^n s_p(f_j) = \\ &= \frac{1}{n} \sum_{j=1}^n \left(\log_\kappa(f_j) - \tilde{\mathbb{E}}_p[(\log_\kappa(f_j))] \right) - \left(\log_\kappa(p) - \tilde{\mathbb{E}}_p[\log_\kappa(p)] \right). \end{aligned} \quad (23)$$

The mean value computed in the reference q is $\bar{s}_q$ and the difference is

$$\bar{s}_p - \bar{s}_q = \log_\kappa(q) - \log_\kappa(p) - \frac{1}{n} \sum_{j=1}^n \tilde{\mathbb{E}}_p[\log_\kappa(f_j)] + \frac{1}{n} \sum_{j=1}^n \tilde{\mathbb{E}}_q[\log_\kappa(f_j)].$$

Hence,

$$\bar{s}_p + \log_\kappa(p) = \bar{s}_q + \log_\kappa(q) + \text{constant}.$$

The probability function is the same in both cases. In fact

$$\begin{aligned} \exp_\kappa(\bar{s}_p - K_p(\bar{s}_p) + \log_\kappa(p)) &= \\ \exp_\kappa(\bar{s}_q - K_p(\bar{s}_p) + \log_\kappa(q) + \text{constant}) &= \\ \exp_\kappa(\bar{s}_q - K_q(\bar{s}_q) + \log_\kappa(q)) & \end{aligned}$$

because of the uniqueness of the normalizing constant.

In conclusion, the probability function

$$\bar{f} = \exp_\kappa(\bar{s}_p - K_p(\bar{s}_p) + \log_\kappa(p)),$$

with $\bar{s}_p$ as in eq. (23) does not depend on the reference p . It is the *barycentre* of the given data points.

The displacement of each data point f_j from the barycentre can be evaluated with any of the two Kaniadakis divergences, either $\tilde{D}(f_j|\bar{f})$ or $\tilde{D}(\bar{f}|f_j)$.

3. Data Analysis

3.1. Kaniadakis divergence

First, we compute the Kaniadakis divergence defined in equation (8). Each point in (i, j) in Figure (1) is the Kaniadakis divergence of (CoDa point) corresponding to the year in the i^{th} row with respect to the (CoDa point) for the year in the j^{th} column. For example, the Kaniadakis divergence between 2008 and 2009 $\tilde{D}(2008|2009) = 0.14$. Most values are smaller than one, except when the reference distribution corresponds to 2008 or 2009 for the most recent years. The year 2009 deviates significantly from other years.

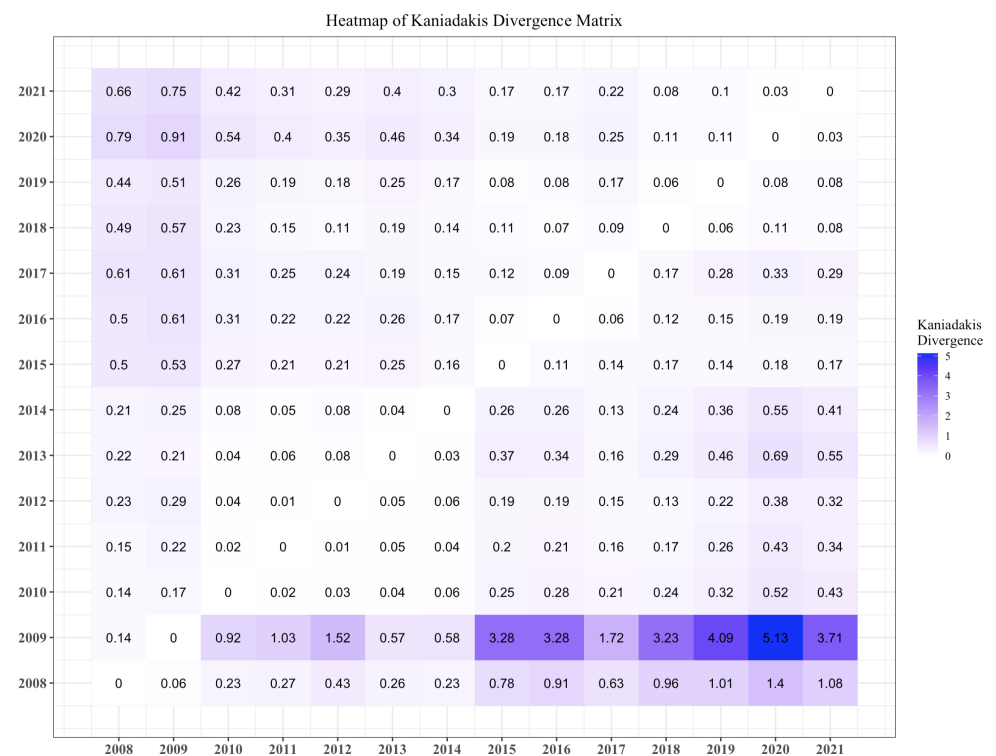


Figure 1. Kaniadakis Divergence on Compositional Data

3.2. Mixture Displacement

The equation (21) calculates the mixture displacement. From Figure (2), the mixture displacement for Greece and Spain is very high for all the years. The value for Spain in 2009 was less than zero—the only negative value for Spain. On the contrary, all other countries do not have too many high values.

The equation (23) calculates the mean value. After determining the mean, we compute the mixture displacement using the mean as a reference. We check that our values abruptly go from -10 to 10. However, the results for Greece and Spain decrease when the mixture displacement from the mean is calculated.

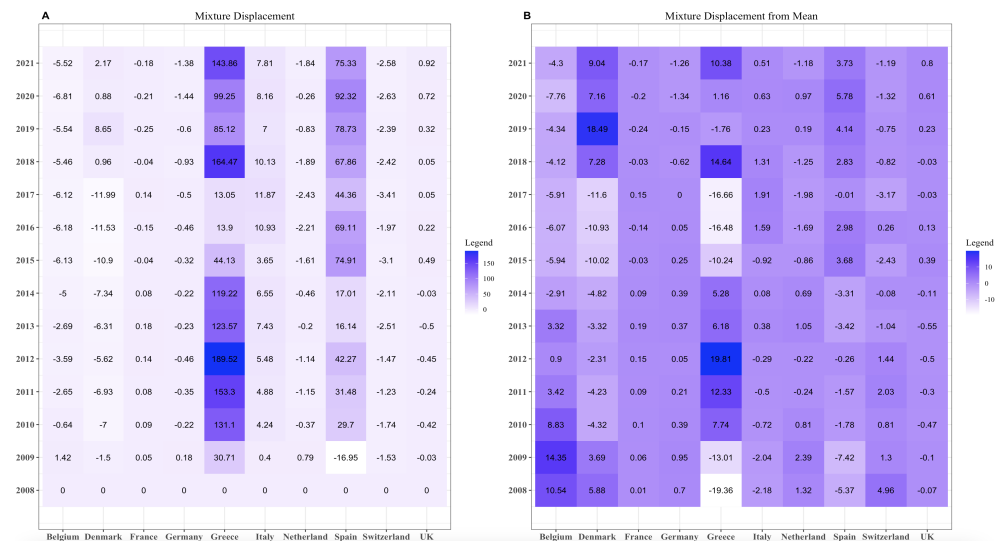


Figure 2. A. Mixture Displacement on Compositional Data by taking 2008 as a reference and B. Mixture Displacement on Compositional Data by taking barycenter as reference

3.3. Exponential Displacement

The equation (22) computes the exponential displacement. Figure (3) is the empirical result of equation (22). Like mixture displacement, while calculating exponential displacement, we can see that Spain and Greece have higher displacement than other European countries. The value for Spain in the year 2009 is meager. Exponential displacement ranges from 0 to -60 when we calculate the exponential displacement from the mean. The only significant changes are for the nations of Spain and Greece, where our values for 2009 for Spain decreased by about 18 times, and for Greece, our values decreased significantly.

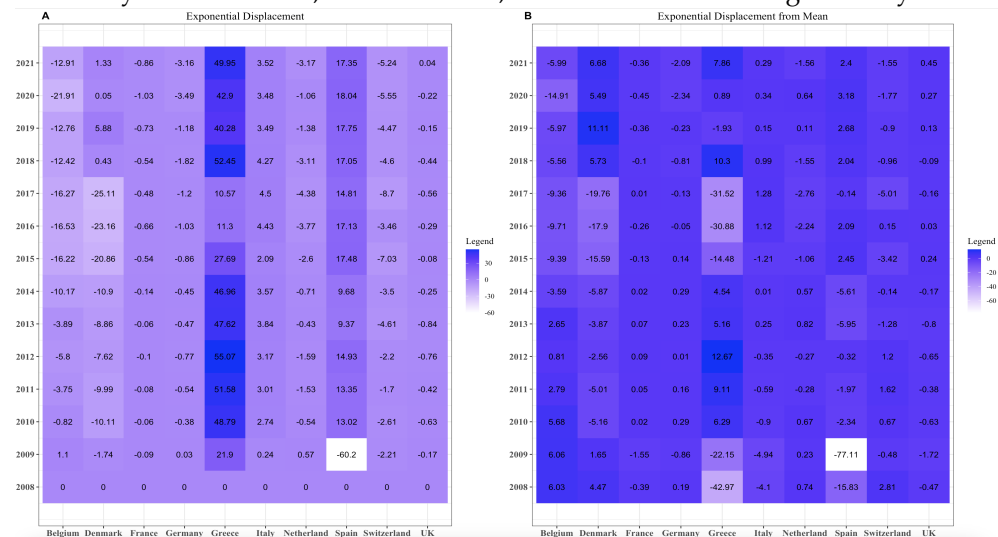


Figure 3. A. Exponential Displacement on Compositional Data by taking 2008 as a reference and B. Exponential Displacement on Compositional Data by taking barycenter as reference

4. Conclusion

In this research, we used compositional data and a particular type of divergence called Kaniadakis divergence. The dataset being examined spans the years 2008 through 2021. First, we build a theoretical framework for Kaniadakis divergence, mixture displacement, and exponential displacement.

The first section of the research establishes the mathematical framework for determining divergence and displacement. Second, we apply those mathematical algorithms to compositional data. The results obtained in the previous section tell us that the countries like

Spain and Greece have more fluctuations when compared to the other European countries. 221
From mixture and exponential displacement, we see that Spain and Greece faced some 222
financial crises compared to other countries. In 2009, the displacement values for Spain 223
and Greece were very different from the other European countries. 224
From the results, we can conclude that the Kaniadakis logarithm will reduce the algebraic 225
computations and can benefit the scientific community that wants accurate monitoring 226
of systemic risk. Using Kaniadakis logarithm, mixture, and exponential displacement 227
on compositional data will be considered a broader research area that tries to reduce the 228
complexity for multiple purposes compared to traditional methods. 229

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